B1GMB42 Software Manual

IndianaDell workspace

2026-07-03

B1GMB42 Software Manual

Machine: Dell Precision Tower 5810 (B1GMB42)
Hostname: Tower5810
OS: Ubuntu 26.04 LTS (resolute)
Workspace: `~/Documents/IndianaDell`

Companion hardware manual: `B1GMB42-slot-port-inventory.md` (slots, GPUs, storage, PERC, ports)

This manual documents every **host-facing install** the IndianaDell workspace provides: apt packages, rustup, Python venvs, built tools, Flatpak apps, GNOME preferences, Plymouth themes, and optional GPU/ROCm tooling. Each chapter covers one topic using the same structure:

1. What gets installed
2. How it is installed
3. How to verify
4. How to customize
5. What `bin/rebuild-machine` does and does not do

Build PDF: `bin/build-software-manual` from the workspace root.

Quick reference: `docs/features-available.md` (cheat sheet, not a replacement for this manual).

Supersedes: flat `B1GMB42-software-inventory.md` (now a stub with links here).

Chapter 1 — Introduction

What this workspace installs

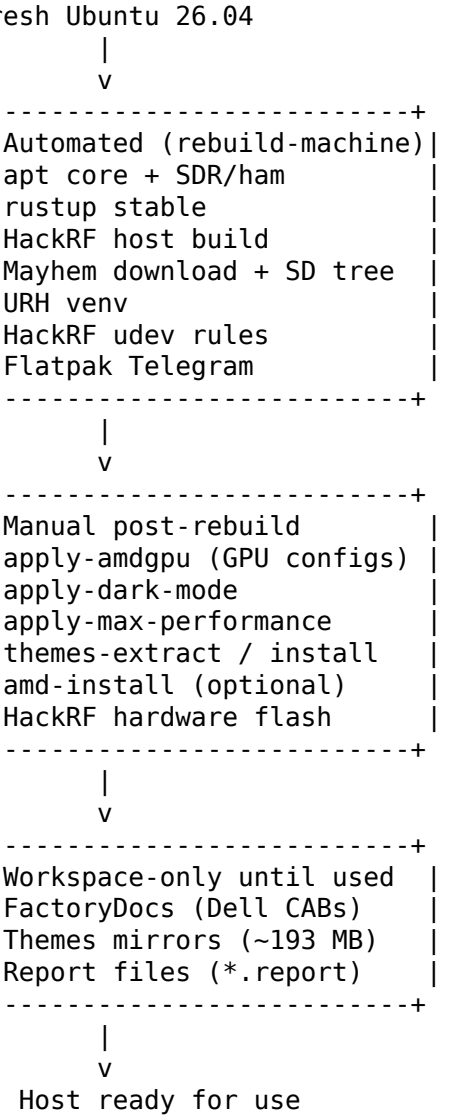
IndianaDell is a **software restoration toolkit** for Tower5810 after a fresh Ubuntu 26.04 install. It does not partition disks, configure ZFS, flash BIOS, or install Windows. It restores the development, SDR, ham

radio, HackRF/Mayhem, documentation, and workstation utility stack documented in the chapters that follow.

The workspace lives at `~/Documents/IndianaDell`. Copy or clone it before running `bin/rebuild-machine`.

Install layering

Software arrives in three layers. Understanding the order prevents skipped steps after a reinstall.



Automated steps run via `bin/rebuild-machine` (see Chapter 2). **Manual** steps are intentional: GPU session files, GNOME gsettings, Plymouth overlay, and hardware flashing need user context or sudo at the right time. **Workspace-only** content stays in the repo until you invoke the matching `bin/` launcher.

Workspace vs host paths

Location	Role
`~/Documents/IndianaDell/`	Source of truth for scripts, themes, HackRF assets
`/usr/`	Apt-installed binaries, Plymouth themes, udev rules (after apply)
`~/.cargo/`	Rust toolchain (rustup)
`hackrf/venv-urh/`	Universal Radio Hacker Python venv
`hackrf/build/`	HackRF host tools built from source
`hackrf/local/`	Optional CMAKE_INSTALL_PREFIX for built libhackrf
`Themes/*/mirror/`	Frozen copies of apt-owned theme files
`FactoryDocs/`	Dell vendor packages (not auto-installed to host)

Reading guide

If you need…	Read
Full restore after reinstall	Ch. 2 + Ch. 3
Python, Rust, pandoc	Ch. 4
Boot/login/desktop look	Ch. 5 + Ch. 7
FirePro GPUs, ROCm	Ch. 6
GNU Radio, gqrx, SoapySDR	Ch. 8
fldigi, WSJT-X, CHIRP	Ch. 9
HackRF, Mayhem, URH	Ch. 10
Telegram	Ch. 11
iotest, dellmerge	Ch. 12
Dell driver CABs	Ch. 13
Known gaps	Ch. 14

If you need...	Read
All `bin/` commands	Appendix A
All apt package names	Appendix B

Related documents

- **Hardware:** `B1GMB42-slot-port-inventory.md` — GPUs, PERC, bays, ports
- **Themes deep-dive:** `Themes/README.md` and per-folder READMEs (27 files)
- **HackRF inventory:** `hackrf/MANIFEST.txt`
- **Apt snapshots:** `apt-full-manifest.txt`, `apt-hamradio-dev-manifest.txt`
- **Rebuild log:** `scripts/rebuild/last-run.log`

Chapter 2 — Rebuild and Recovery

What gets installed

`bin/rebuild-machine` restores the full automated software stack in one run (~15–30 minutes, network dependent). It installs **90 apt packages** from `scripts/rebuild/package-lists.sh` (APT_CORE 37 + APT_SDR_HAM 53), plus rustup, HackRF repos/build, Mayhem v2.4.0 assets, URH venv, udev rules, and Flatpak Telegram.

How it is installed

```
cd ~/Documents/IndianaDell
chmod +x bin/* scripts/rebuild/*.sh
bin/rebuild-machine                  # full restore
bin/rebuild-machine --verify-only    # check only, no installs
```

Environment overrides:

Variable	Effect
`SKIP_TELEGRAM=1`	Skip Flatpak Telegram install
`SKIP_HACKRF_BUILD=1`	Skip cmake build of HackRF host tools

Phases (from `scripts/rebuild/rebuild-machine.sh`):

Phase	Action
1	`apt-get update`
2	Install `APT_CORE` (37 packages) — build, Python, docs, GPU utils, flatpak
3	Install `APT_SDR_HAM` (53 packages) — GNU Radio, ham, SDR hardware, HackRF
4	Flatpak remote + `org.telegram.desktop` (unless skipped)
5	rustup stable if `rustc` missing
6	Clone HackRF/Mayhem/URH repos under `hackrf/repos/`
7	Build HackRF host tools to `hackrf/build/` (unless skipped)
8	Download Mayhem v2.4.0 + extract SD card tree
9	Create `hackrf/venv-urh/` with URH
10	Install HackRF udev rules; chmod `bin/` and scripts
11	Regenerate apt manifests; run `verify_stack`

Debconf preseed: `xastir/install-setuid` is set to `false` before apt to avoid interactive hangs.

Log file: `scripts/rebuild/last-run.log`

How to verify

```
bin/rebuild-machine --verify-only
```

`verify_stack` checks:

- Every package in `APT_CORE` and `APT_SDR_HAM` via `dpkg-query`
- Commands: `rustc`, `cargo`, `gnuradio-config-info`, `grcc`, `gqrx`, `fldigi`, `wsjtx`, `chirpw`, `hackrf_info`, `inspectrum`, `pandoc`, `xelatex`, `vkcube`
- `hackrf/venv-urh/bin/urh`
- Mayhem firmware zip and extracted SD tree
- Built `hackrf/build/hackrf-tools/src/hackrf_sweep`
- Launchers: `dellmerge`, `gpu-stress`, `iotest`, `apply-amdgpu`, `rebuild-machine`
- Flatpak Telegram (unless `SKIP_TELEGRAM=1`)

Exit code 0 means all checks passed.

How to customize

- **Add apt packages:** Edit `scripts/rebuild/package-lists.sh`, update Appendix B, re-run rebuild.
- **Pin HackRF/Mayhem version:** Edit `hackrf/scripts/download-mayhem.sh` and MANIFEST; rebuild does not auto-upgrade pinned releases.
- **Skip heavy steps:** Use SKIP_* env vars for CI or partial recovery.

What rebuild does / does not do

Rebuild **does**	Rebuild **does not**
apt install all listed packages	Partition disks or ZFS
rustup, HackRF build, Mayhem download	`sudo bin/apply-amdgpu`
URH venv, udev rules	`bin/apply-dark-mode` / `apply-max-performance`
Regenerate `apt-full-manifest.txt`	Plymouth theme install
chmod workspace scripts	Flash HackRF / PortaPack firmware
	`bin/amd-install` (ROCm)
	Install FactoryDocs CABs to Windows

After a successful rebuild, continue with **Chapter 3 — Post-Rebuild Checklist**.

Chapter 3 — Post-Rebuild Checklist

`bin/rebuild-machine` intentionally stops before steps that need a logged-in desktop, a reboot, or hardware attached. Run this checklist once per fresh install.

1. GPU session configuration

Tower5810 has three AMD FirePro cards (W5000/W5100). Multi-GPU Wayland/X11 configs live in `etc/`.

```
cd ~/Documents/IndianaDell
sudo bin/apply-amdgpu
sudo reboot
```

Verify after reboot: `echo $WAYLAND_DISPLAY`, `glxinfo -B`, `vkcube` on each display if needed.

See Chapter 6 for ROCm (`bin/amd-install`) — optional and not supported for ML on these GPUs.

2. GNOME session preferences

Run as the **desktop user** (not root):

```
bin/apply-dark-mode          # Yaru-dark GTK, shell, icons, GDM greeter
bin/apply-max-performance    # no suspend, dimming, or night light
```

Verify:

```
gsettings get org.gnome.desktop.interface color-scheme
powerprofilesctl get
```

See Chapter 7 for every gsettings key touched.

3. Boot splash (optional)

Default Ubuntu **bgrt** Plymouth theme shows Dell BGRT center + Ubuntu watermark. To customize:

```
bin/themes-extract                    # refresh mirrors + extract logos
# edit Themes/boot/overlay/watermark.png or background.png
sudo bin/themes-install-boot          # or --oem / --no-watermark
sudo reboot
```

Restore factory: `sudo bin/themes-restore-boot`

See Chapter 5.

4. HackRF hardware (when device is available)

```
source bin/hackrf-env
hackrf_info                           # should list board
bin/hackrf-flash-mayhem               # extract USB flash bundle
# follow bundle README for DFU flash
bin/hackrf-prepare-sdcard             # ensure SD tree is extracted
# copy hackrf/sd-card/mayhem-v2.4.0/* to FAT32 microSD root
```

Verify: `hackrf_info`, on-device Mayhem version, SD apps visible on PortaPack.

See Chapter 10.

5. Documentation PDFs

```
bin/build-software-manual             # this manual
pandoc B1GMB42-slot-port-inventory.md -o B1GMB42-slot-port-inventory.pdf \
  --pdf-engine=xelatex -V mainfont="Noto Sans" -V monofont="DejaVu Sans Mono"
```

6. Machine inventory baseline

```
bin/dellmerge > b1gmb42.report
sudo bin/iotest                          # optional storage survey
bin/gpu-stress 60 vkcube                 # optional GPU smoke test
```

7. FactoryDocs recovery (optional)

Only 19 of 101 pre-crash Dell packages are on disk. Re-download per `FactoryDocs/README.md` and `MANIFEST-pre-crash.txt`. These are **workspace archives**, not installed by rebuild.

See Chapter 13.

Quick verification block

```
cd ~/Documents/IndianaDell
bin/rebuild-machine --verify-only
source bin/hackrf-env
. ~/.cargo/env && rustc --version
gnuradio-config-info --version
bin/urh --version
```

Summary table

Step	Command	Reboot?
GPU configs	`sudo bin/apply-amdgpu`	Yes
Dark mode	`bin/apply-dark-mode`	No
Max performance	`bin/apply-max-performance`	No
Custom boot	`sudo bin/themes-install-boot`	Yes
HackRF flash	`bin/hackrf-flash-mayhem` + DFU	Maybe
ROCm (optional)	`bin/amd-install`	Yes
Manual PDF	`bin/build-software-manual`	No

Chapter 4 — Development Toolchain

What gets installed

Component	Version / path	Install source
Python	3.14 (system)	Ubuntu base + apt

Component	Version / path	Install source
pip, venv	apt	`python3-pip`, `python3-venv`
numpy, scipy, matplotlib	apt	science stack
PyQt5	apt	GNU Radio Companion, some tools
Rust	1.96.1 stable	rustup to `~/.cargo/bin`
C/C++ build	apt	`build-essential`, `cmake`, `pkg-config`
Clang/LLVM	apt	bindgen-style Rust, native tooling
SSL/USB/FFTW/Volk	apt dev libs	SDR and native builds
ARM cross-compile	apt	PortaPack Mayhem firmware (`gcc-arm-none-eabi`)
pandoc + XeLaTeX	apt	Manual PDF generation
Git, curl, wget	apt	repos and downloads

Python bindings verified on this host: `gnuradio`, `SoapySDR`, `Hamlib` (capital H in Python).

Project venv: URH lives in `hackrf/venv-urh/` (Chapter 10), not system-wide.

How it is installed

Apt (rebuild Phase 2): packages listed in Appendix B under “Development” and shared dev libraries.

Rust (rebuild Phase 5):

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh -s -- -y --default-
source ~/.cargo/env
```

Rebuild skips rustup if `rustc` is already on PATH.

How to verify

```
python3 --version
python3 -c "import numpy, scipy, matplotlib; print('OK')"
. ~/.cargo/env && rustc --version && cargo --version
cmake --version | head -1
arm-none-eabi-gcc --version | head -1
```

```
pandoc --version | head -1
xelatex --version | head -1
```

How to customize

- **New Python project venv:** `python3 -m venv myproject/.venv && . myproject/.venv/bin/activate`
- **Rust toolchain:** `rustup toolchain install stable`, `rustup component add clippy`
- **Per-project Rust SDR crates:** `cargo add rtlsdr` etc. — no workspace scaffold ships with IndianaDell
- **Manual PDF:** `bin/build-software-manual` or pandoc on any `.md` in the repo

What rebuild does / does not do

Does	Does not
Install all dev apt packages	Create application-specific venvs beyond URH
Install rustup if missing	Pin a non-stable Rust toolchain
chmod workspace scripts	Install IDE or editor plugins

Chapter 5 — Themes (Boot, Login, Desktop)

What gets installed

The **Themes/** module (~194 MB with mirrors) documents and customizes three visual stages:

Stage	What you see	Apt packages	Workspace folder
Boot	Dell BGRT center + spinner + Ubuntu watermark	`plymouth`, `plymouth-theme-spinner`, …	`Themes/boot/`
Login	GDM on GNOME Shell	`gdm3`, `gnome-shell`, `yaru-theme-gnome-shell`	`Themes/login/`
Desktop	Yaru GTK, icons, shell	`yaru-theme-gtk`, `yaru-theme-icon`, …	`Themes/desktop/`

Active Plymouth theme: `bgrt` at `/usr/share/plymouth/themes/bgrt/bgrt.plymouth`

Extracted boot logos:

- `Themes/boot/extracted/bgrt-firmware-oem.png` — Dell from UEFI BGRT (`/sys/firmware/acpi/bgrt/image`)
- `Themes/boot/extracted/ubuntu-watermark-dark.png` — bottom Ubuntu text

Custom Plymouth install target: `indianadell` theme under `/usr/share/plymouth/themes/indian`

Each subfolder has its own `README.md` and `apt-packages.txt`. See `Themes/MANIFEST.txt` for a one-page map.

How it is installed

Themes are **not** applied by `bin/rebuild-machine`. Use launchers:

```
bin/themes-extract              # snapshot apt-owned files + extract logos (~193 M
sudo bin/themes-install-boot    # install custom Plymouth from boot/overlay/
sudo bin/themes-restore-boot    # revert to stock bgrt
bin/apply-dark-mode             # login + desktop dark (Chapter 7)
```

Boot overlay workflow:

```
cp my-logo.png Themes/boot/overlay/watermark.png    # bottom Ubuntu text only
sudo bin/themes-install-boot

cp my-splash.png Themes/boot/overlay/background.png
sudo bin/themes-install-boot --oem                  # replace center (disable BGRT

sudo bin/themes-install-boot --no-watermark         # hide bottom logo
```

Every Plymouth install runs `update-initramfs -u` — **reboot required** to preview.

How to verify

```
plymouth-set-default-theme -l
plymouth-set-default-theme                    # shows active theme
ls Themes/boot/extracted/
gsettings get org.gnome.desktop.interface gtk-theme   # after apply-dark-mode
```

How to customize

Deep documentation lives in module READMEs — this chapter summarizes:

- **Boot:** `Themes/boot/README.md` — overlay/, stock/, indianadell/ plymouth definition

- **Login:** `Themes/login/README.md` — GDM greeter assets mirrored from apt
- **Desktop:** `Themes/desktop/README.md` — GTK/icon/shell Yaru variants

Center Dell logo comes from BIOS BGRT firmware, not an Ubuntu file. Change it in BIOS setup or use `--oem` with your PNG.

What rebuild does / does not do

Does	Does not
Install `plymouth`, `gdm3`, Yaru via apt (Phase 2–3)	Run `themes-extract` or `themes-install-boot` Set dark mode or GDM greeter prefs Copy overlay PNGs into initramfs

Run Chapter 3 checklist items 2 and 3 after rebuild.

Chapter 6 — GPU and Display

What gets installed

Component	Source	Purpose
`vulkan-tools`, `mesa-utils`, `clinfo`	apt (APT_CORE)	Vulkan/OpenGL/OpenCL diagnostics
`etc/` multi-GPU configs	workspace	Wayland, X11, udev, GDM tweaks
`amd-radeon/` scripts	workspace	Optional ROCm driver install
`bin/gpu-stress`	workspace	3-GPU Vulkan smoke test

Hardware (this machine): 2x AMD FirePro W5000 + 1x FirePro W5100. Vulkan and OpenCL work for graphics/compute smoke tests. **ROCm ML/HIP is not supported** on these cards (see Chapter 14).

How it is installed

Apt (automated): GPU utility packages install during rebuild Phase 2.

Session configs (manual):

```
sudo bin/apply-amdgpu    # runs etc/apply.sh
sudo reboot
```

`etc/apply.sh` installs:

- `etc/environment.d/99-amdgpu-wayland.conf`
- `etc/X11/xorg.conf.d/20-amdgpu-multi-gpu.conf`
- `etc/modprobe.d/amdgpu-multigpu.conf`
- `etc/udev/rules.d/99-amdgpu-multigpu.rules`
- `etc/profile.d/amdgpu-multigpu.sh`
- `etc/gdm3/custom.conf` (if present)

Optional ROCm:

```
bin/amd-preflight        # check prerequisites
bin/amd-install          # full driver stack from amd-radeon/
bin/amd-verify
bin/amd-uninstall        # remove if needed
```

How to verify

```
vkcube                   # Vulkan cube (per display)
clinfo | head -30        # OpenCL platforms/devices
glxinfo -B               # OpenGL renderer
bin/gpu-stress 60 vkcube # stress all GPUs ~60s
lspci -nn | grep -i vga
```

How to customize

- Edit files under `etc/` before re-running `sudo bin/apply-amdgpu`
- ROCm install scripts and README in `amd-radeon/` — machine-specific; read preflight output
- Hardware details: `B1GMB42-slot-port-inventory.md` Video section

What rebuild does / does not do

Does	Does not
Install mesa-utils, vulkan-tools, clinfo	Run `apply-amdgpu`
Ensure `bin/gpu-stress` is executable	Install ROCm
	Configure monitor layout (use GNOME Settings)

Required post-rebuild: `sudo bin/apply-amdgpu` + reboot (Chapter 3).

Chapter 7 — GNOME Session

What gets installed

GNOME desktop preferences applied via gsettings (user session) and optionally GDM (login greeter). No apt packages beyond what Ubuntu desktop already provides (gdm3, gnome-shell, Yaru themes).

Scripts:

- scripts/gnome/apply-dark-mode.sh via bin/apply-dark-mode
- scripts/gnome/apply-max-performance.sh via bin/apply-max-performance

How it is installed

Run as the **logged-in desktop user**, not root:

```
bin/apply-dark-mode
bin/apply-max-performance
```

apply-dark-mode

Sets:

Schema	Key	Value
org.gnome.desktop.interface	color-scheme	prefer-dark
org.gnome.desktop.interface	gtk-theme	Yaru-dark (override: GTK_THEME=)
org.gnome.desktop.interface	icon-theme	Yaru-dark
org.gnome.shell.ubuntu	color-scheme	prefer-dark
org.gnome.desktop.wm.preferences	theme	Yaru-dark
org.gnome.settings-daemon.plugins.color	night-light	disabled

With APPLY_GDM=1 (default), also sets GDM greeter dark via sudo dbus-run-session gsettings ….

apply-max-performance

Sets:

- Power plugin: no suspend on AC/battery, no idle dim, lid close does nothing
- Session idle delay: 0
- Screensaver: no blanking or lock on idle
- Night light: off

- `powerprofilesctl set performance` when available

How to verify

```
gsettings get org.gnome.desktop.interface color-scheme
gsettings get org.gnome.desktop.interface gtk-theme
gsettings get org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type
powerprofilesctl get
```

Log out and back in to confirm GDM greeter if dark mode was applied.

How to customize

- Override themes: `GTK_THEME=Yaru-viridian bin/apply-dark-mode`
- Skip GDM: `APPLY_GDM=0 bin/apply-dark-mode`
- Revert individual keys with `gsettings reset …` or GNOME Settings app
- Login/desktop asset mirrors: `Themes/login/`, `Themes/desktop/`

What rebuild does / does not do

Does	Does not
Install gdm3, gnome-shell, Yaru via apt	Change any gsettings Set performance power profile

Run both launchers after every fresh install (Chapter 3).

Chapter 8 — GNU Radio and Desktop SDR

What gets installed

Component	Version	Packages / path
GNU Radio	3.10.12.0	`gnuradio`, `gnuradio-dev`, `gnuradio-doc`
Companion blocks	apt	`gr-osmosdr`, `gr-limesdr`, `gr-fosphor`, `gr-air-modes`, `gr-hpsdr`, `gr-dab`, `gr-satellites`
SoapySDR	apt + Python	`libsoapysdr-dev`, `python3-soapysdr`, modules

Component	Version	Packages / path
Hardware libs	apt	RTL-SDR, HackRF, Airspy, bladeRF, Lime, UHD
Desktop apps	apt	`gqrx-sdr`, `quisk`, `inspectrum`, `hacktv`

SoapySDR modules on this host: HackRF, RTL-SDR (osmosdr), Airspy, bladeRF, Lime, MiriSDR, HydraSDR, PlutoSDR, Red Pitaya, remote, audio, UHD.

How it is installed

All packages in `APT_SDR_HAM` (rebuild Phase 3). Dev libraries in `APT_CORE` support building OOT modules.

Typical workflow:

```
grcc myflowgraph.grc          # compile Companion graph
gqrx                          # general receiver GUI
quisk                         # transceiver GUI
inspectrum capture.cf32       # visualize IQ files
```

For HackRF-specific host tools and URH, see Chapter 10.

How to verify

```
gnuradio-config-info --version
grcc --help | head -1
python3 -c "import gnuradio; print(gnuradio.__version__)"
python3 -c "import SoapySDR; print('SoapySDR OK')"
SoapySDRUtil --info
gqrx --version 2>/dev/null || command -v gqrx
```

With hardware attached:

```
rtl_test -t                   # RTL-SDR
hackrf_info                   # HackRF
```

How to customize

- Add OOT modules: `sudo apt install gr-<name>` or build from source against `gnuradio-dev`
- GPU waterfall: `gr-fosphor` blocks in flowgraphs
- Filtered apt list: `apt-hamradio-dev-manifest.txt` (178 SDR/ham-related packages on full system)

What rebuild does / does not do

Does	Does not
Install full GNU Radio + gr-* stack	Calibrate specific SDR hardware
Install gqrx, quisk, inspectrum	Install SDRangel or SigDigger
Verify `gnuradio-config-info`, `grcc`, `gqrx` in verify_stack	Flash firmware on SDR devices

Chapter 9 — Ham Radio (Desktop)

What gets installed

Application	Command	Apt package	Role
fldigi	`fldigi`	`fldigi`	Digital modes (PSK, RTTY, …)
WSJT-X	`wsjtx`	`wsjtx`, `wsjtx-data`	FT8, JT65, weak-signal
CHIRP	`chirpw`, `chirpc`	`chirp`	Radio programming
direwolf	`direwolf`	`direwolf`	Sound-card TNC / APRS
gpredict	`gpredict`	`gpredict`	Satellite pass prediction
grig	`grig`	`grig`	Hamlib rig control GUI
xastir	`xastir`	`xastir`, `xastir-data`	APRS map client
Hamlib	API	`libhamlib-dev`, `libhamlib-utils`, `python3-hamlib`	Rig control library

How it is installed

Apt packages in APT_SDR_HAM (rebuild Phase 3).

xastir debconf: rebuild preseeds `xastir/install-setuid boolean false` to avoid interactive install hangs.

```
fldigi &
wsjtx &
chirpw &
direwolf -p
gpredict &
grig &
xastir &
```

How to verify

```
command -v fldigi wsjtx chirpw direwolf gpredict grig xastir
python3 -c "import Hamlib; print('Hamlib OK')"
bin/rebuild-machine --verify-only   # checks fldigi, wsjtx, chirpw
```

Configure rig control in each app via Hamlib model selection.

How to customize

- Radio definitions: CHIRP stock configs + your radio CSV
- WSJT-X: `~/.config/WSJT-X/`
- xastir maps: `xastir-data` package + user map sources
- direwolf: `~/.direwolf/direwolf.conf`

What rebuild does / does not do

Does	Does not
Install all ham desktop apps + Hamlib	Configure radios or call signs
Preseed xastir setuid prompt	Set up APRS IS or igates
Verify fldigi, wsjtx, chirpw commands	Install fldigi/WSJT-X from source

Chapter 10 — HackRF and PortaPack Mayhem

What gets installed

Recommended firmware: PortaPack Mayhem v2.4.0

Apt packages

`hackrf`, `hackrf-firmware`, `libhackrf-dev`, `hackrf-doc`, `inspectrum`, `hacktv`, `dfu-util`, `openocd`, ARM GCC toolchain, plus GNU Radio/SoapySDR deps (Chapter 8).

Built from source (`hackrf/build/`)

Tool	Notes
`hackrf_sweep`	Spectrum sweep — not in older apt splits
`hackrf_info`, `hackrf_transfer`, ...	Newer libhackrf (0.10.0) than apt alone
`libhackrf.so`	Under `hackrf/build/libhackrf/src/`

Install prefix: `hackrf/local/` (CMAKE_INSTALL_PREFIX).

Release assets (`hackrf/releases/`)

File	Size	Purpose
`FIRMWARE_mayhem_v2.4.0.zip`	8.4 MB	USB flash bundle
`COPY_TO_SDCARD_hackrf_mayhem_v2.4.0/`	201 MB	PortaPack microSD
`no-world-map.zip`		
`OCI_hackrf_mayhem_2.4.0.ppfw.tar`	2.2 MB	Web flasher image

Extracted SD tree: `hackrf/sd-card/mayhem-v2.4.0/` (276 MB, 84 apps in APPS/)

Source repos (`hackrf/repos/`)

`hackrf`, `mayhem-firmware` (+ submodules), `portapack-hackrf`, `urh`, `hacktv`

Python venv

`hackrf/venv-urh/` — Universal Radio Hacker **2.10.0** (PyQt6). Launch: `bin/urh`

udev

`hackrf/scripts/99-hackrf.rules` installed to `/etc/udev/rules.d/` — plugdev access.

How it is installed

Automated (rebuild Phases 6–10):

1. Clone repos (shallow, skip if `.git` exists)
2. Init Mayhem submodules if needed
3. cmake + build HackRF host (skip: `SKIP_HACKRF_BUILD=1`)
4. `hackrf/scripts/download-mayhem.sh`
5. `hackrf/scripts/prepare-sdcard.sh`
6. URH venv if missing
7. `hackrf/scripts/setup-udev.sh`

Manual (hardware present):

```
source bin/hackrf-env
bin/hackrf-flash-mayhem         # extract flash bundle
bin/hackrf-prepare-sdcard       # re-extract SD payload
bin/hackrf-build-mayhem         # compile Mayhem from source (advanced)
bin/hackrf-download-mayhem      # re-fetch releases
```

PATH: `source bin/hackrf-env` adds `hackrf/build/hackrf-tools/src` and `hackrf/local/bin`.

How to verify

```
bin/rebuild-machine --verify-only
source bin/hackrf-env
hackrf_info                     # needs USB device
hackrf/build/hackrf-tools/src/hackrf_sweep --help | head -1
bin/urh --version
ls hackrf/sd-card/mayhem-v2.4.0/APPS | wc -l
```

How to customize

- **Upgrade Mayhem:** Edit `hackrf/scripts/download-mayhem.sh` version URLs; re-run download + prepare
- **SD apps:** Copy subsets from `sd-card/mayhem-v2.4.0/APPS/` to microSD
- **Full inventory:** `hackrf/MANIFEST.txt`

Mayhem v2.4.0 highlights

On-device: Morse RX/TX, RTTY, FPV detect, ADSB, ACARS, BLE, TPMS, KeeLoq, EPIRB, SAME, MDC-1200, P25, KISS TNC, Looking Glass, SubGHz, Flipper TX, waterfall designer, time sink.

SD apps include: `fpv_detect`, `kiss_tnc`, `keeloqtx`, `siggen`, `fmradio`, `sstvrx`, `wardrivemap`, `waterfall_designer`, and more.

What rebuild does / does not do

Does	Does not
Clone repos, build tools, download Mayhem	DFU-flash firmware to hardware
Create URH venv, install udev	Format or write microSD in a reader
Verify zip, SD tree, hackrf_sweep	Test with HackRF USB attached

Chapter 11 — Flatpak Applications

What gets installed

App	Flatpak ID	Version (this host)
Telegram	`org.telegram.desktop`	6.9.3

Runtime dependency: `flatpak` package from `APT_CORE`.

How it is installed

Rebuild Phase 4:

```
flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.fla
flatpak install -y flathub org.telegram.desktop
```

Skip with `SKIP_TELEGRAM=1 bin/rebuild-machine`.

How to verify

```
flatpak list --app | grep telegram
flatpak run org.telegram.desktop --version 2>/dev/null || true
bin/rebuild-machine --verify-only
```

How to customize

```
flatpak update org.telegram.desktop
flatpak override --user org.telegram.desktop …   # permissions, env
```

`bin/apply-dark-mode` sets `prefer-dark` color scheme; Flatpak GTK4 apps pick this up via portal when supported.

What rebuild does / does not do

Does	Does not
Install `flatpak` via apt	Install SDRangel, SigDigger (not on Flathub)
Add flathub remote + Telegram	Pin Telegram to a specific commit
Treat Telegram miss as non-fatal on install (warns in log)	Install other Flatpak apps by default

Chapter 12 — Machine Utilities

What gets installed

Workspace scripts for inventory, storage benchmark, and GPU stress — no dedicated apt packages beyond shared GPU utils (`mesa-utils`, `vulkan-tools`).

Utility	Launcher	Script	Output
Dell inventory	`bin/dellmerge`	`scripts/dell/dellmerge.sh`	`dellmerge/` `*.report` files
Storage survey	`bin/iotest`	`scripts/storage/iotest.sh`	I/O tests (sudo)
GPU stress	`bin/gpu-stress`	`scripts/gpu/gpu-stress.sh`	Vulkan/EGL per GPU

Example reports in workspace: `b1gmb42.report`, `B1GMB42.ioperf` (from prior runs).

How it is installed

Scripts ship with the workspace. Rebuild Phase 10 runs `chmod +x` on `bin/*` and `scripts/*/*.sh`.

```
bin/dellmerge > b1gmb42.report
sudo bin/iotest
bin/gpu-stress 60 vkcube
```

How to verify

```
[[ -x bin/dellmerge && -x bin/iotest && -x bin/gpu-stress ]] && echo OK
bin/rebuild-machine --verify-only   # checks dellmerge, gpu-stress, iotest, apply
head -20 b1gmb42.report 2>/dev/null || bin/dellmerge | head -20
```

How to customize

- Edit `scripts/dell/dellmerge.sh` to add inventory fields
- `gpu-stress` accepts duration and backend (vkcube default)
- `iotest` targets block devices — read script header before running on production pools

What rebuild does / does not do

Does	Does not
chmod utility launchers	Run dellmerge or iotest automatically
Verify launcher executables exist	Archive reports to a fixed path

Chapter 13 — FactoryDocs (Workspace Archive)

What gets installed

Nothing on the Linux host automatically. FactoryDocs is a sorted archive of Dell T5810 vendor support packages for Windows recovery, firmware, and reference — stored only in the workspace.

Metric	Pre-crash	Current
Sorted packages	101	**19**
GPU drivers (FirePro/Quadro)	Yes	**Missing**
PERC H710 driver/firmware	Yes	**Missing**
Audio / input drivers	Yes	**Missing**
Win7/10/WinPE CAB packs	Yes	**Missing**

Full pre-crash file list: `FactoryDocs/MANIFEST-pre-crash.txt` (91 items flagged `MISS`).

Layout

Folder	Contents
`System-T5810/`	BIOS, chipset, ME, TPM, manuals
`GPU/`	AMD FirePro, NVIDIA (**empty — re-download**)
`Storage/`	PERC H710, Intel RST, SSD/HDD firmware
`Network/`	Intel Ethernet
`Dell-Management/`	Command Update, Configure
`Expansion-Cards/`	Serial, Thunderbolt docs
`_Misc/`	Non-Dell packages
`_incoming/`	Drop new Dell downloads here

How it is installed

Ingest new downloads:

```
cp ~/Downloads/* ~/Documents/IndianaDell/FactoryDocs/_incoming/
python3 ~/Documents/IndianaDell/FactoryDocs/_sort_factory_docs.py
```

Priority re-downloads from Dell T5810 drivers:

1. `GPU/AMD-FirePro/Windows/` — Video_Driver_C5FPW (W5000/W5100)
2. `Storage/RAID-Controller-PERC/Windows/` — PERC H710
3. `System-T5810/Windows-10/` — T5810-win10 CAB
4. `Audio/Windows/` — Audio_Driver_5P33P
5. `Dell-Management/Windows/` — Command Configure, Monitor

Windows installs use Dell CAB/EXE packages from these folders — not apt.

How to verify

```
find FactoryDocs -type f ! -path '*/_incoming/*' | wc -l
cat FactoryDocs/README.md
grep MISS FactoryDocs/MANIFEST-pre-crash.txt | wc -l
```

How to customize

- `_sort_factory_docs.py` dedupes and sorts by hardware category
- Cross-reference hardware manual: `B1GMB42-slot-port-inventory.md` for what hardware needs drivers

What rebuild does / does not do

Does	Does not
Nothing	Copy FactoryDocs to system paths Install Windows drivers Download missing CABs

FactoryDocs recovery is a **manual** ongoing task (Chapter 3 item 7).

Chapter 14 — Gaps and Limits

Documented boundaries of what IndianaDell does **not** install or support on this host.

Not installed

Item	Notes
SDRangel, SigDigger	Not on Flathub; use gqrx + URH + inspectrum
Rust SDR crate workspace	Toolchain only — add crates per project with `cargo add`
ZFS / disk layout tools	Out of scope — handled at OS install time
Windows / dual-boot	FactoryDocs holds CABs; no auto-install
HackRF hardware test	No device attached at last verify (2026-07-03)

Lost in TPM/ZFS crash

Item	Recovery
Pre-crash full apt list	**Not found** — use `apt-full-manifest.txt` from current rebuild
Pre-crash FactoryDocs	82/101 packages missing — see `MANIFEST-pre-crash.txt`
Pre-crash apt-hamradio list	Regenerated by rebuild `save_manifests()`

GPU / ROCm matrix

Capability	W5000 / W5100 on Ubuntu 26.04
Desktop amdgpu	Yes — with `etc/` configs
Vulkan (`vkcube`)	Yes
OpenCL (`clinfo`)	Yes (limited)
ROCm HIP / ML training	**No** — not in AMD ROCm support matrix for FirePro W5000/W5100

Use `bin/amd-preflight` before `bin/amd-install`; expect warnings for these GPUs.

Rebuild intentional omissions

These remain **manual** by design (see Chapter 3):

- `sudo bin/apply-amdgpu`
- `bin/apply-dark-mode`, `bin/apply-max-performance`
- `bin/themes-extract`, `sudo bin/themes-install-boot`
- `bin/hackrf-flash-mayhem` (DFU with hardware)
- `bin/amd-install` (optional ROCm)

Future enhancement could fold GNOME/theme steps into rebuild; current manual documents the gap explicitly.

Encryption / TPM

Hardware manual recommends **not** re-enabling ZFS encryption until TPM + recovery strategy is documented. IndianaDell rebuild does not touch encryption.

When something fails

1. Read `scripts/rebuild/last-run.log`
2. Run `bin/rebuild-machine --verify-only` for targeted MISS lines
3. Re-run individual phases (apt, HackRF build, Mayhem download) manually
4. Check chapter-specific verify sections

Reporting issues

Capture:

```
bin/dellmerge > debug.report
bin/rebuild-machine --verify-only 2>&1 | tee verify.log
uname -a && lsb_release -a
```

Appendix A — bin/ Launchers

All launchers live in `~/Documents/IndianaDell/bin/`. Add to PATH optionally:

```
export PATH="$HOME/Documents/IndianaDell/bin:$PATH"
```

Launcher	Runs	Chapter
`rebuild-machine`	`scripts/rebuild/rebuild-machine.sh`	2
`build-software-manual`	`scripts/docs/build-software-manual.sh`	1
`dellmerge`	`scripts/dell/dellmerge.sh`	12
`gpu-stress`	`scripts/gpu/gpu-stress.sh`	6, 12
`iotest`	`scripts/storage/iotest.sh`	12
`apply-amdgpu`	`etc/apply.sh`	6
`amd-install`	`amd-radeon/install-all.sh`	6
`amd-preflight`	`amd-radeon/00-preflight.sh`	6
`amd-verify`	`amd-radeon/04-verify.sh`	6
`amd-uninstall`	`amd-radeon/uninstall.sh`	6
`apply-dark-mode`	`scripts/gnome/apply-dark-mode.sh`	5, 7
`apply-max-performance`	`scripts/gnome/apply-max-performance.sh`	7
`themes-extract`	`Themes/scripts/extract-all.sh`	5
`themes-install-boot`	`Themes/scripts/install-boot-theme.sh`	5
`themes-restore-boot`	`Themes/scripts/install-boot-theme.sh --restore-stock`	5
`hackrf-env`	`sources hackrf/scripts/env.sh`	10

Launcher	Runs	Chapter
`urh`	`hackrf/scripts/launch-urh.sh`	10
`hackrf-setup-udev`	`hackrf/scripts/setup-udev.sh`	10
`hackrf-download-mayhem`	`hackrf/scripts/download-mayhem.sh`	10
`hackrf-prepare-sdcard`	`hackrf/scripts/prepare-sdcard.sh`	10
`hackrf-flash-mayhem`	`hackrf/scripts/flash-mayhem.sh`	10
`hackrf-build-mayhem`	`hackrf/scripts/build-mayhem.sh`	10

Note: `hackrf-env` must be **sourced**, not executed: `source bin/hackrf-env`

Sudo required: `apply-amdgpu`, `themes-install-boot`, `themes-restore-boot`, `iotest`, `hackrf-setup-udev` (udev install), `amd-install`

Appendix B — Apt Packages by Chapter

Source of truth: `scripts/rebuild/package-lists.sh` (`APT_CORE` + `APT_SDR_HAM`).
Total: 90 packages installed by `bin/rebuild-machine` (37 core + 53 SDR/ham).
Full system snapshot: `apt-full-manifest.txt` (~2257 packages after rebuild).
SDR/ham filter: `apt-hamradio-dev-manifest.txt` (~178 related packages).

Packages below are grouped by manual chapter. Shared dev libraries appear under Chapter 4 and are reused by Chapters 8–10.

Chapter 4 — Development

`build-essential`, `cmake`, `pkg-config`, `git`, `curl`, `wget`, `unzip`, `python3-pip`, `python3-venv`, `python3-dev`, `python3-numpy`, `python3-scipy`, `python3-matplotlib`, `python3-yaml`, `python3-requests`, `python3-pyqt5`, `python3-psutil`, `libssl-dev`, `clang`, `llvm-dev`, `libclang-dev`, `libusb-1.0-0-dev`, `libfftw3-dev`, `libvolk-dev`, `portaudio19-dev`, `libsndfile1-dev`, `libboost-dev`, `libboost-program-options-dev`, `pandoc`, `texlive-latex-recommended`,

texlive-fonts-recommended, texlive-xetex

Chapter 6 — GPU and Display

vulkan-tools, mesa-utils, mesa-utils-bin, clinfo

Chapter 8 — GNU Radio and SDR

gnuradio, gnuradio-dev, gnuradio-doc, gr-osmosdr, gr-limesdr, gr-fosphor, gr-air-modes, gr-hpsdr, gr-dab, gr-satellites, libsoapysdr-dev, python3-soapysdr, soapysdr-module-osmosdr, soapysdr-module-mirisdr, uhd-soapysdr, rtl-sdr, librtlsdr-dev, airspy, libairspy-dev, bladerf, libbladerf-dev, limesuite, limesuite-udev, uhd-host, libuhd-dev, gqrx-sdr, quisk, inspectrum, hacktv

Chapter 9 — Ham Radio

libhamlib-dev, libhamlib-utils, python3-hamlib, fldigi, wsjtx, wsjtx-data, chirp, direwolf, gpredict, grig, xastir, xastir-data

Chapter 10 — HackRF and Mayhem

hackrf, hackrf-firmware, libhackrf-dev, hackrf-doc, dfu-util, openocd, gcc-arm-none-eabi, binutils-arm-none-eabi, libnewlib-arm-none-eabi, ccache, lz4, bzip2

(Also uses Chapter 8 packages for GNU Radio/SoapySDR integration.)

Chapter 11 — Flatpak

flatpak (application org.telegram.desktop installed via flatpak, not apt)

Chapters 5, 7, 12, 13 — No dedicated apt arrays

Themes, GNOME prefs, machine utilities, and FactoryDocs use workspace scripts or Ubuntu desktop packages already on the base install (gdm3, gnome-shell, plymouth, Yaru themes) — not enumerated separately in package-lists.sh.